

Researcher and engineer with first-author published work on explaining spiking neural networks, and five years building machine-learning systems that run in production: agentic LLM systems and knowledge graphs at BIIE, edge computer vision on factory lines at Siemens, neuromorphic and event-based perception at Intel Labs.

EXPERIENCE

BIIE.ai, Zürich · Data Science Manager and ML Engineer, AI System Owner Aug 2025 – present

- Own the design and build of a multi-agent LLM system for scientific peer review, from ingestion to synthesis, now in production use by scientists across the institute.
- Built the knowledge graph underneath it end to end: custom ontology decomposing manuscripts into hypotheses, methods, experiments, results and claims, plus ingestion, entity resolution, store and multi-hop traversal for provenance and claim verification.
- Built the evaluation framework (rubric-based LLM-as-judge scoring, critique matching, inter-rater reliability) and a benchmark of preprint, review and published-version triplets; designed and ran a blinded, pre-registered validation study on 30 preprints with a panel of 16 PhD domain experts.

Siemens AG, Vienna · Full-Stack ML Engineer, ML, MLOps and Data Science Apr 2023 – May 2025

- Developed and deployed edge computer vision (detection, segmentation, anomaly detection) on industrial PCs and Nvidia Jetson with HMI and robot integration, across customer projects including solar-panel quality inspection and multi-sensor inspection along a 200 m line.
- Co-led foundation-model research for industrial anomaly detection and defect analysis in a small research team, including fine-tuning SAM with LoRA; researched diffusion models for synthetic defect generation.
- Co-built and debugged the distributed training setup behind it: 16 H100 GPUs on a multi-node Slurm cluster with DDP and DeepSpeed. Standardized the team's data lake on AWS, MinIO and Delta Lake, preparing roughly 20 TB of image data.

Intel Labs, Munich · ML and Neuromorphic Research Scientist, Dependability and Robustness for Cyber-Physical Systems Sep 2020 – Apr 2022

- Developed gradient-based feature-attribution methods for backpropagation-trained spiking neural networks, adapting Saliency and Integrated Gradients to event-camera and frame data; published as first author. Trained the networks from scratch with surrogate gradients and ran models on Intel Loihi.
- Showed on CIFAR-10 and DVS128-Gesture, across neural codings, that under 5% of spikes drive a full change in prediction, at two to three orders of magnitude less compute than model-agnostic methods.
- Built event-camera vehicle detection for the Providentia++ motorway-sensing programme, including DVS-to-RGB co-calibration and sensor fusion, and fine-tuned YOLO for night and heavy precipitation.

SKILLS

Code	Python, C++, SQL
ML and training	PyTorch, TensorFlow, Hugging Face Transformers, scikit-learn, LoRA, surrogate gradients, multi-node Slurm with DDP and DeepSpeed
LLMs and agents	LangGraph, MCP, OpenAI, Anthropic and Gemini APIs, retrieval-augmented generation, LLM-as-judge evaluation
Knowledge	Knowledge graphs, ontology design, entity resolution, multi-hop traversal
Vision and edge	OpenCV, YOLO, event-based vision, TensorRT, ONNX, Triton, DeepStream, Jetson Linux, Docker, CUDA, AWS
Robotics	ROS and ROS2, Gazebo, CARLA, CoppeliaSim, SLAM

EDUCATION

MSc Artificial Intelligence and Machine Learning
Maastricht University, 2022

BSc Computer Science, minors in Biology and Neuroscience
Université de Fribourg, 2018 Thesis: loop closure for SLAM using a spiking winner-take-all network.

PUBLICATIONS AND LANGUAGES

Gradient-based feature-attribution explainability methods for spiking neural networks. Bitar, Rosales, Paulitsch. Frontiers in Neuroscience 17, 2023. doi:10.3389/fnins.2023.1153999

An agentic system for scientific peer review: architecture, evaluation framework, and blinded human validation. Bitar et al. Working title, in preparation.

German, French, Spanish and English, all fluent. Arabic, intermediate.